Activity 3C: Motion Alarm

Purpose: Learn how to create a circuit that uses light, sound, and motion to detect moving objects. Use a servo motor to add a moving pop-up to animate your alarm.

Materials:

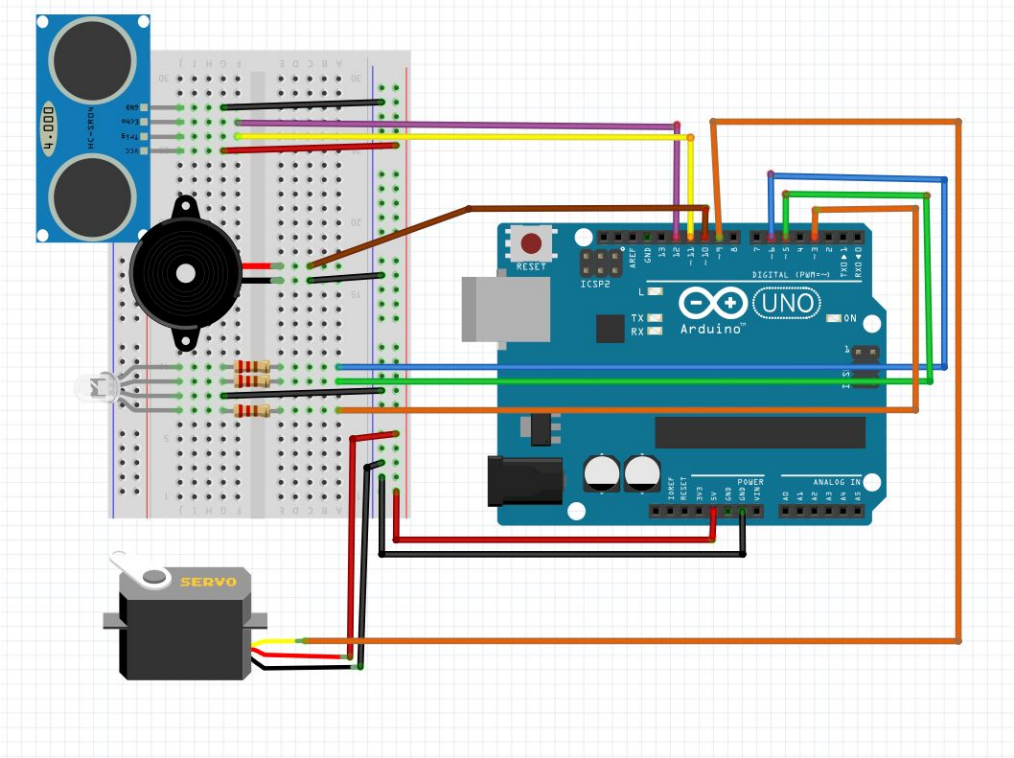
RGB LED, three 330 or 220 Ω, distance sensor, 10 jumper wires, servo motor, piezzo speaker

New Components/Concepts:

Mechanisms: This circuit gets really fun when you start to use your servo to animate the world around you. To do this, you'll need to connect your servo to some physical mechanisms. Tape and small screws are easy ways to connect things to your servo. You can also loop a paper clip through the small holes in the servo arm to serve as a linkage.

Procedure:

1. Wire the project using the following schematic.



2. You can create your code in the BlocklyDuino Editor then paste it to the Codebender website. Click BlocklyDuino Reference for a link to all the blockly codes. Use the following blocks to create your code.

   a. Configure the ultrasound like you did in your last lesson

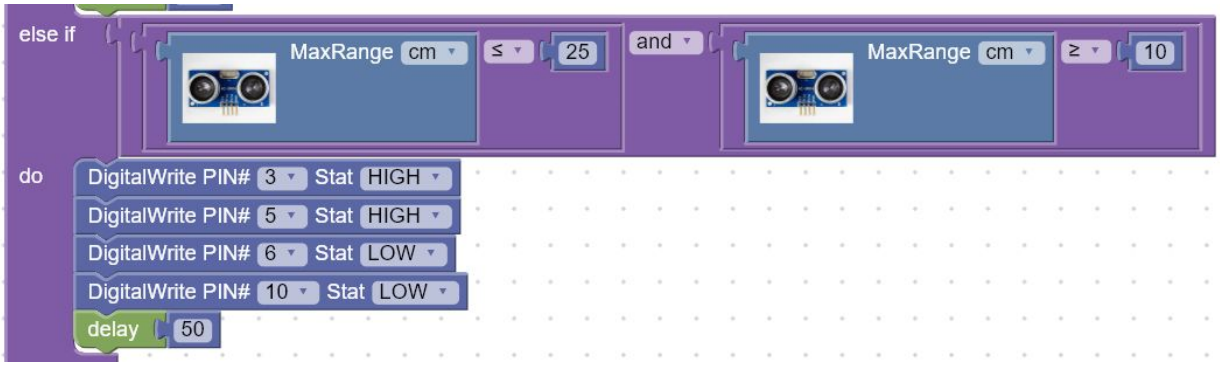


   b. Now let's set up the alarm for less than 10 cm. We want the LED to turn red and "waggle" the servo from 45 to 135 degrees.



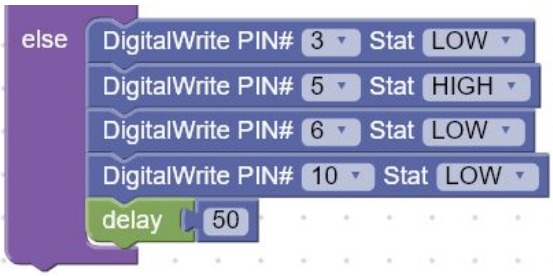
   c. Now let's set up the alarm to warn the intruder between 10 and 25 cm. This will turn on the yellow light and make sure the speaker alarm is off.

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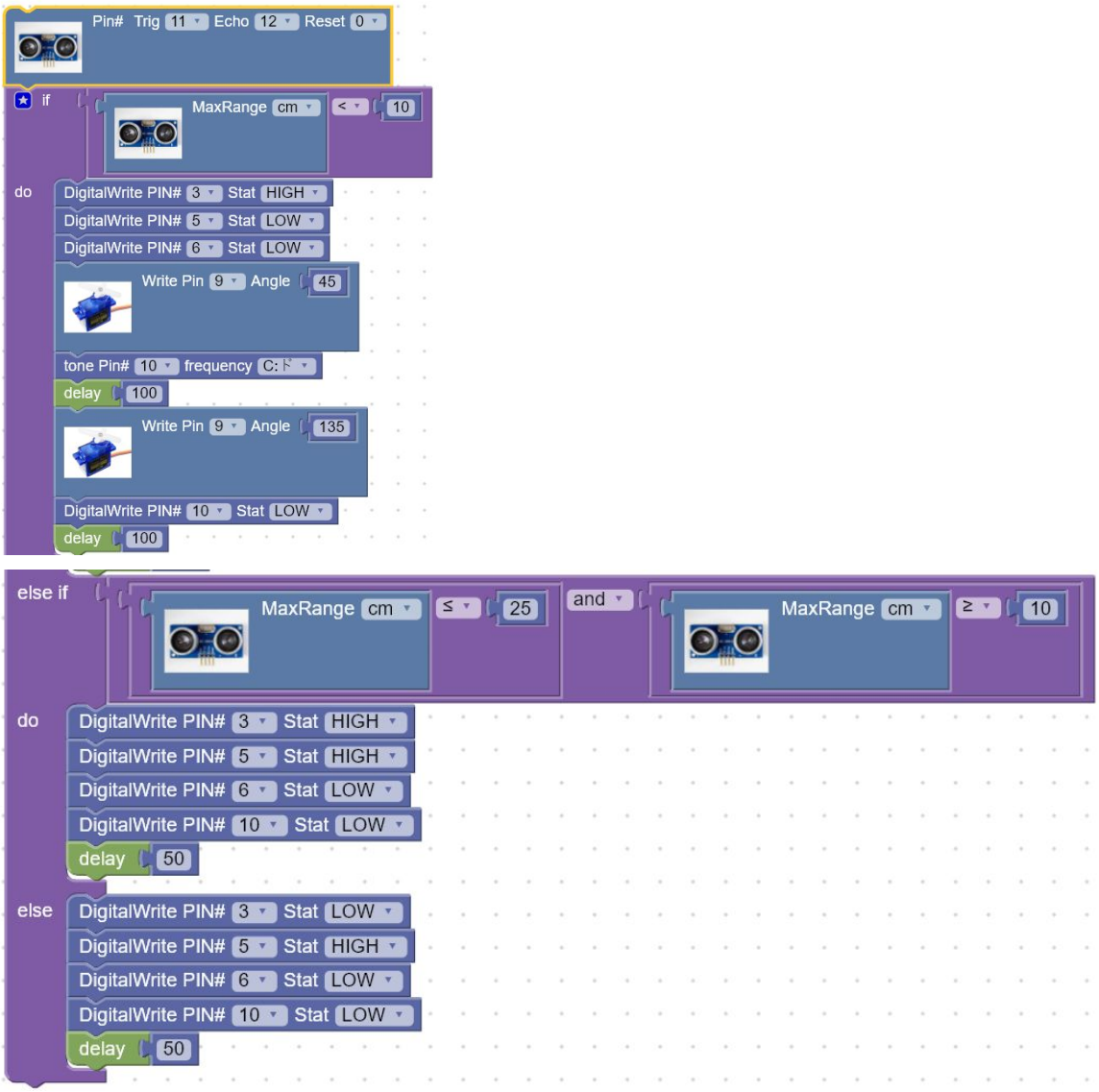


d. Lastly, let's turn on the green light for all other distances, to signify "all is clear"

3. Your completed code should look like this.



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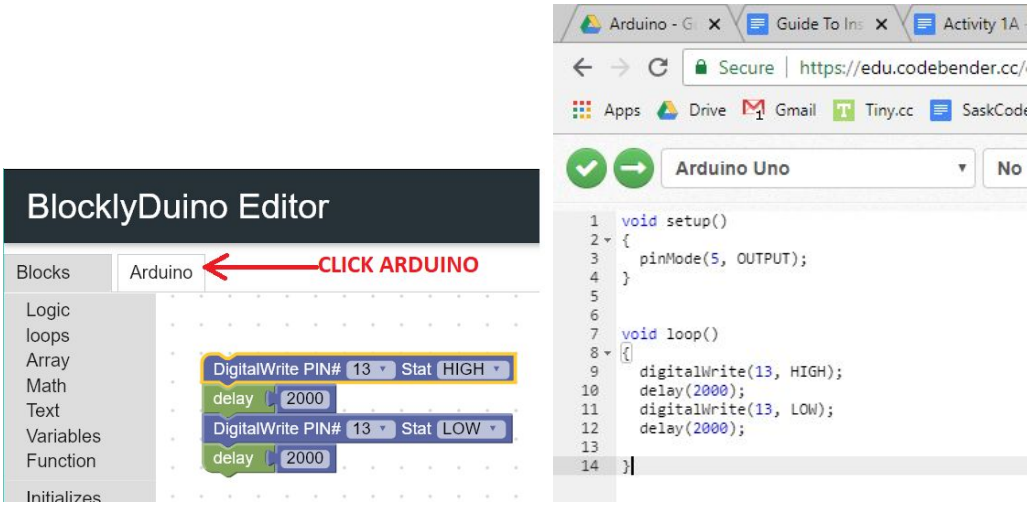


*****SAVE your sketch to your BlocklyDuino Folder that you created*****

5. Save your sketch as “3C-Motion Alarm.”

6. Copy and paste the Arduino Code from BlocklyDuino into the Codebender website. Hook up your arduino to your computer using the USB cable.

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6. Upload your program to the arduino by clicking the arrow button inside the green circle.

What you should see:

The RGB LED will behave as in your last circuit. It will be green when objects are far, yellow when they are midrange and red when they are close. When an object is close, the buzzer will also beep, and the servo will rotate back and forth. If you decided to attach a pop-up, it will move back and forth.

Check what distance the sensor is reading.

1. If the distance is less than 10 cm, make the RGB LED red. Then make the servo rotate back and forth and make the buzzer beep.
2. If the distance is between 10 and 20 cm, make the RGB LED yellow.
3. If the distance value is not equal to the fist two conditions, make the RGB LED green.

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Assignment 3C: Motion Alarm.

1. **CHANGE THE SERVO BEHAVIOR:** Try changing the way your servo behaves when the distance sensor is triggered. Which part of the code do you have to change to achieve this?

   Note: Its best to avoid driving the servo all the way to 0 or 180 degrees. It can cause the motor to jitter, which is bad for the servo. So typically a range of 20-170 works best.

2. **CHANGE THE ALARM SETTINGS:** Try altering the code so the alarm goes off from much farther or closer distances. What do you have to change in the code to do this?

3. **Add a buzzer.** Below are the blocks that you will need to add to your code to add an alarm buzzer. They are found under the In/Out menu in the sub-menu titled, "Others." Good Luck. The buzzer was included in the schematic that you wired and should be wired to pin 10.



   a. These blocks will play a frequency when the alarm is triggered. Where in your code should you put these blocks?
   b. Change the frequencies and duration to change your alarm sounds. You can even add more blocks to add more frequencies.

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